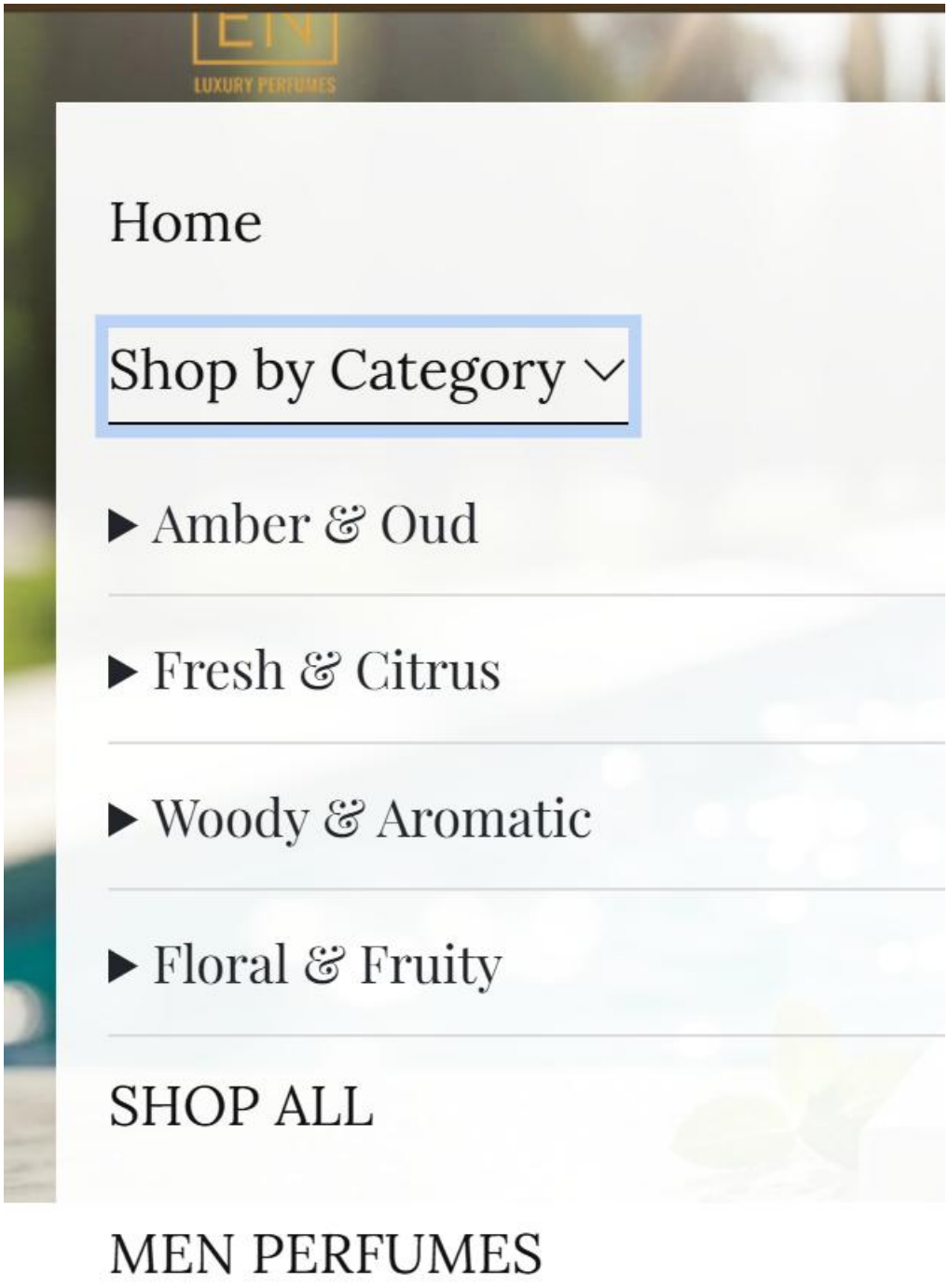
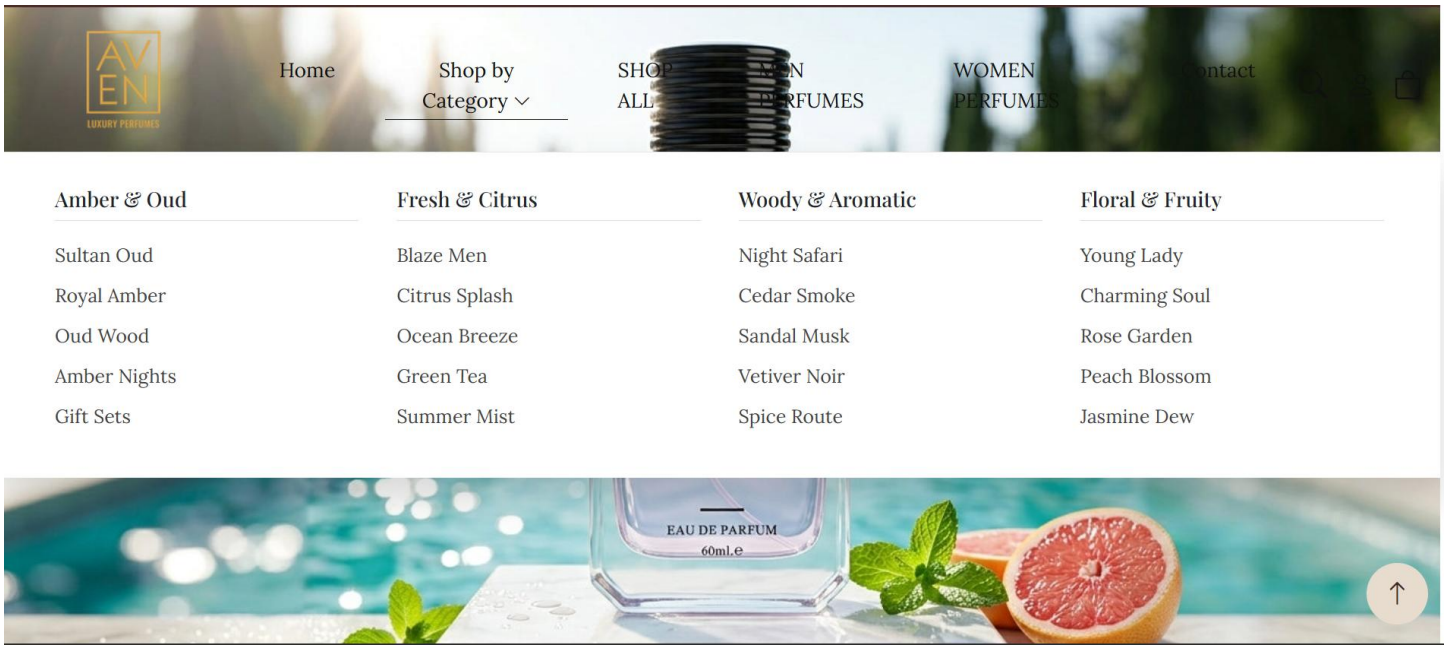
Humaira Akram Sheikh
Registration number: FA24-BCS-044
midterm — Aven Perfumes (N=44, BP=992px, K=4)

Task 1 — Mega-menu

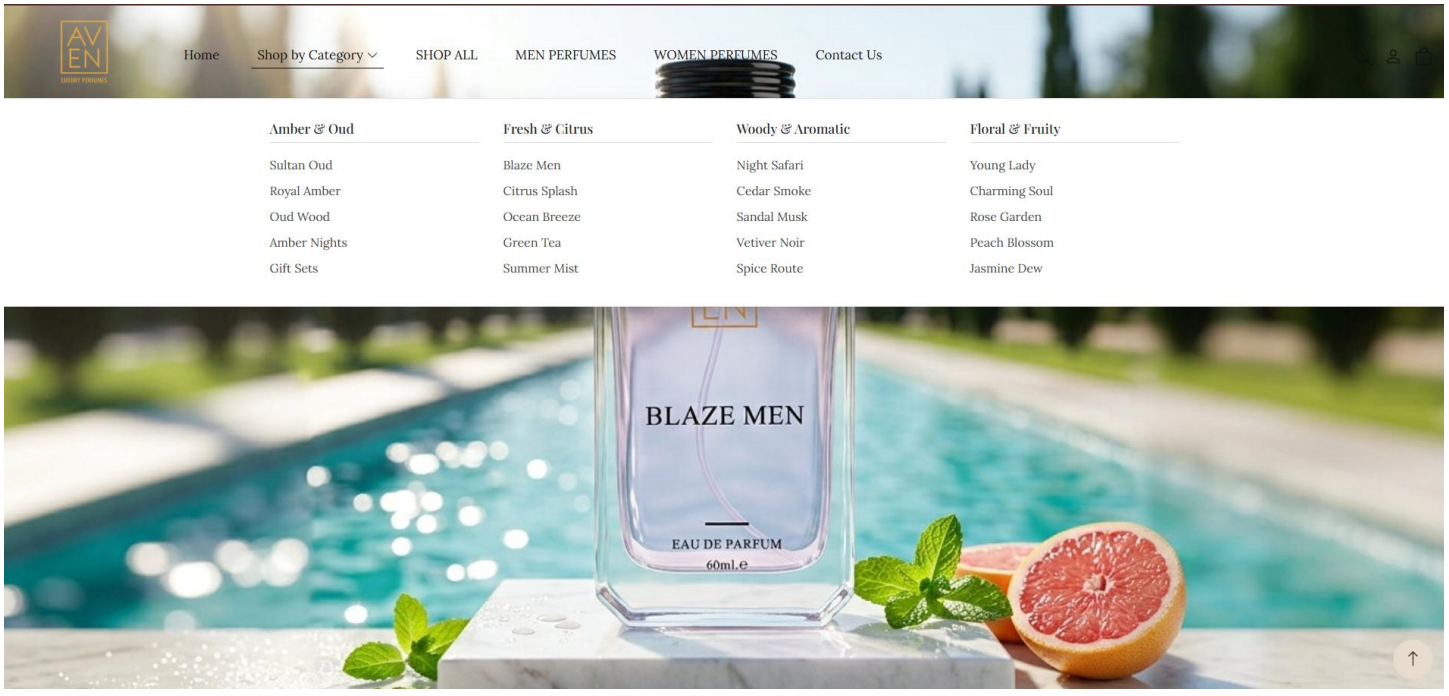
Screenshots



Mega-menu / accordion — 375px



Mega-menu / accordion — 992px



Mega-menu / accordion — 1280px

HTML (navbar item)

```
<li class="nav-item mega">
            <button type="button" class="nav-link mega-toggle" aria-haspopup="true">Shop by Category <i
class="bi bi-chevron-down small"></i></button>
          <!-- TASK 1: desktop mega panel (shown from BP=992px up, opened by :hover / :focus-within) -->
          <div class="mega-panel" id="megaPanel">
            <div class="mega-cols">
               <section class="mega-col">
                 <h3>Amber &amp; Oud</h3>
                 <ul>
                   <li><a href="#">Sultan Oud</a></li>
                   <li><a href="#">Royal Amber</a></li>
                   <li><a href="#">Oud Wood</a></li>
                   <li><a href="#">Amber Nights</a></li>
                   <li><a href="#">Gift Sets</a></li>
                 </ul>
               </section>
```

```
        <section class="mega-col">
          <h3>Fresh &amp; Citrus</h3>
          <ul>
            <li><a href="#">Blaze Men</a></li>
            <li><a href="#">Citrus Splash</a></li>
            <li><a href="#">Ocean Breeze</a></li>
            <li><a href="#">Green Tea</a></li>
            <li><a href="#">Summer Mist</a></li>
          </ul>
        </section>
        <section class="mega-col">
          <h3>Woody &amp; Aromatic</h3>
          <ul>
            <li><a href="#">Night Safari</a></li>
            <li><a href="#">Cedar Smoke</a></li>
            <li><a href="#">Sandal Musk</a></li>
            <li><a href="#">Vetiver Noir</a></li>
            <li><a href="#">Spice Route</a></li>
          </ul>
        </section>
        <section class="mega-col">
          <h3>Floral &amp; Fruity</h3>
          <ul>
            <li><a href="#">Young Lady</a></li>
            <li><a href="#">Charming Soul</a></li>
            <li><a href="#">Rose Garden</a></li>
            <li><a href="#">Peach Blossom</a></li>
            <li><a href="#">Jasmine Dew</a></li>
          </ul>
        </section>
      </div>
    </div>
    <!-- TASK 1: mobile accordion (shown below BP) -->
    <div class="mega-acc">
      <details>
        <summary>Amber &amp; Oud</summary>
        <ul>
          <li><a href="#">Sultan Oud</a></li>
          <li><a href="#">Royal Amber</a></li>
          <li><a href="#">Oud Wood</a></li>
          <li><a href="#">Amber Nights</a></li>
          <li><a href="#">Gift Sets</a></li>
        </ul>
      </details>
      <details>
        <summary>Fresh &amp; Citrus</summary>
        <ul>
          <li><a href="#">Blaze Men</a></li>
          <li><a href="#">Citrus Splash</a></li>
          <li><a href="#">Ocean Breeze</a></li>
          <li><a href="#">Green Tea</a></li>
          <li><a href="#">Summer Mist</a></li>
        </ul>
      </details>
      <details>
        <summary>Woody &amp; Aromatic</summary>
        <ul>
          <li><a href="#">Night Safari</a></li>
          <li><a href="#">Cedar Smoke</a></li>
          <li><a href="#">Sandal Musk</a></li>
          <li><a href="#">Vetiver Noir</a></li>
          <li><a href="#">Spice Route</a></li>
        </ul>
      </details>
      <details>
        <summary>Floral &amp; Fruity</summary>
        <ul>
          <li><a href="#">Young Lady</a></li>
          <li><a href="#">Charming Soul</a></li>
          <li><a href="#">Rose Garden</a></li>
          <li><a href="#">Peach Blossom</a></li>
          <li><a href="#">Jasmine Dew</a></li>
        </ul>
```

```
            </details>
          </div>
          </li>
```

CSS

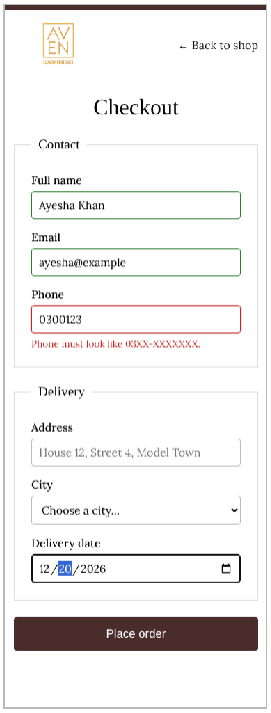
```
/* ========================================================================
   MIDTERM TASK 1 — CSS-only mega-menu            (BP = 992px, K = 4)
   To change BP: edit the two 992px / 991.98px media queries below AND
   Bootstrap's navbar-expand-lg on the <nav> (it collapses at 992px too).
   ======================================================================== */
.navbar { z-index: 2000; }                 /* above grid, cards, sticky/fixed elements */
.mega-toggle { background: none; border: 0; border-bottom: 1px solid transparent; font-family: inherit; }
.mega-toggle:hover, .mega-toggle:focus-visible { border-bottom-color: #111; }

/* default (mobile): the desktop panel is not used, the accordion is */
.mega-panel { display: none; }
.mega-acc details { border-bottom: 1px solid #ddd; }
.mega-acc summary { padding: 10px 0; cursor: pointer; font-family: 'Playfair Display', Georgia, serif; }
.mega-acc ul { list-style: none; padding: 0 0 10px 12px; margin: 0; }
.mega-acc li a { display: block; padding: 4px 0; color: #444; text-decoration: none; }

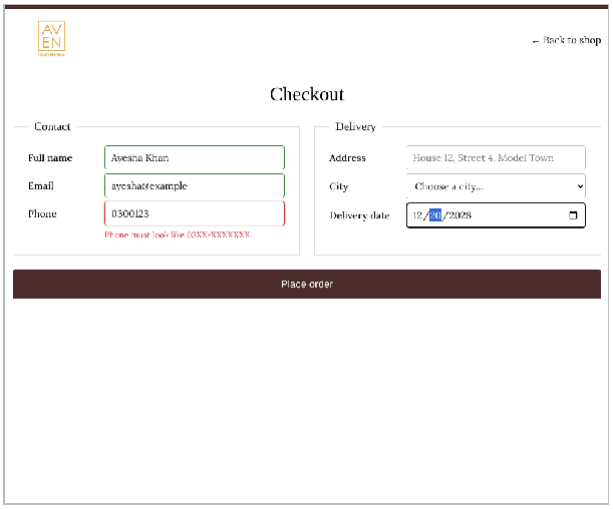
@media (min-width: 992px) {                /* ---- BP and wider----------*/
  .mega-acc { display: none; }
  .mega-panel {
    position: absolute;                    /* out of flow: never pushes content down */
    top: 100%; left: 0; right: 0;          /* full width, directly under the navbar (.navbar is
position:absolute) */
    background: #fff;
    border-top: 1px solid #e6e6e6;
    box-shadow: 0 18px 30px rgba(0, 0, 0, .15);
    padding: 28px 48px 34px;
  }
  /* hover bridge: the menu item's own box grows downward (padding) without moving layout (negative
margin),
     so the pointer never leaves :hover while travelling from "Shop by Category" into the panel */
  .mega { padding-bottom: 60px; margin-bottom: -60px; }
  .mega:hover > .mega-panel,
  .mega:focus-within > .mega-panel { display: block; }    /* hover OR Tab focus; closes when focus leaves */
  .mega-cols { display: grid; grid-template-columns: repeat(4, 1fr); gap: 32px; max-width: 1200px; margin:
0 auto; }
  .mega-col h3 { font-family: 'Playfair Display', Georgia, serif; font-size: 1.1rem; margin: 0 0 12px;
padding-bottom: 8px; border-bottom: 1px solid #ddd; }
  .mega-col ul { list-style: none; margin: 0; padding: 0; }
  .mega-col li a { display: block; padding: 5px 0; color: #444; text-decoration: none; font-size: 1rem; }
  .mega-col li a:hover, .mega-col li a:focus-visible { color: #111; text-decoration: underline; }
}
```

Task 2 — Checkout form

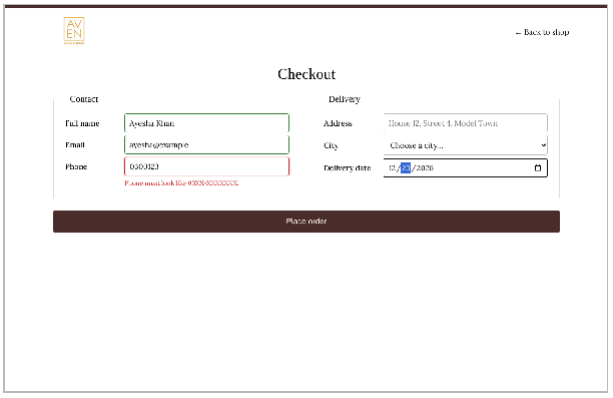
Screenshots (after interacting with some invalid fields)



Checkout form — 375px



Checkout form — 992px



Checkout form — 1280px

checkout.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Checkout — Aven Perfumes</title>
  <link  href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"  rel="stylesheet">
  <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.11.3/font/bootstrap-icons.min.css"
rel="stylesheet">
  <link href="https://fonts.googleapis.com/css2?
family=Lora:wght@400;500&family=Playfair+Display:wght@400;500&display=swap" rel="stylesheet">
  <link href="styles.css" rel="stylesheet">
</head>
<body>
  <div class="top-bar"></div>

  <header class="checkout-header container-xl d-flex align-items-center justify-content-between py-3">
```

```
    <a href="index.html"><img src="images/logo.png" alt="Aven Luxury Perfumes" height="64"></a>
    <a href="index.html" class="checkout-back">&larr; Back to shop</a>
  </header>

  <main class="container-xl checkout">
    <h1 class="text-center mb-4">Checkout</h1>

    <form class="checkout-form" action="#" method="get">

      <fieldset>
        <legend>Contact</legend>

        <div class="field">
          <label for="fullname">Full name</label>
          <input type="text" id="fullname" name="fullname" required minlength="3" autocomplete="name"
placeholder="Ayesha Khan">
          <span class="msg">Please enter your full name (at least 3 characters).</span>
        </div>

        <div class="field">
          <label for="email">Email</label>
          <input type="email" id="email" name="email" required autocomplete="email"
placeholder="name@example.com">
          <span class="msg">Please enter a valid email address.</span>
        </div>

        <div class="field">
          <label for="phone">Phone</label>
          <input type="tel" id="phone" name="phone" required pattern="03[0-9]{2}-[0-9]{7}"
autocomplete="tel" placeholder="0300-1234567" title="Format: 03XX-XXXXXXX">
          <span class="msg">Phone must look like 03XX-XXXXXXX.</span>
        </div>
      </fieldset>

      <fieldset>
        <legend>Delivery</legend>

        <div class="field">
          <label for="address">Address</label>
          <input type="text" id="address" name="address" required minlength="8" autocomplete="street-
address" placeholder="House 12, Street 4, Model Town">
          <span class="msg">Please enter your full street address.</span>
        </div>

        <div class="field">
          <label for="city">City</label>
          <select id="city" name="city" required>
            <option value="">Choose a city…</option>
            <option>Lahore</option>
            <option>Karachi</option>
            <option>Islamabad</option>
            <option>Rawalpindi</option>
            <option>Faisalabad</option>
          </select>
          <span class="msg">Please choose a city.</span>
        </div>

        <div class="field">
          <label for="date">Delivery date</label>
          <!-- min = exam date (2026-10-29); max = exam date + N days (N = 44 -> 2026-12-12) -->
          <input type="date" id="date" name="date" required min="2026-10-29" max="2026-12-12">
          <span class="msg">Pick a date between 29 Oct 2026 and 12 Dec 2026.</span>
        </div>
      </fieldset>

      <button type="submit" class="checkout-submit">Place order</button>
    </form>
  </main>
</body>
</html>
```

CSS

```
/* ========================================================================
   MIDTERM TASK 2 — checkout form (HTML5 validation + CSS only)     BP = 992px
   ======================================================================== */
.checkout { padding-bottom: 80px; max-width: 1100px; }
.checkout h1 { font-family: 'Playfair Display', Georgia, serif; font-weight: 400; }
.checkout-back { color: #111; text-decoration: none; }

.checkout-form { display: grid; grid-template-columns: 1fr; gap: 24px; }       /* below BP: one column */
.checkout-form fieldset { border: 1px solid #ddd; padding: 20px 24px 24px; margin: 0; min-width: 0; }
.checkout-form legend { float: none; width: auto; padding: 0 10px; margin: 0; font-family: 'Playfair
Display', Georgia, serif; font-size: 1.2rem; }

/* inside each fieldset: grid, labels above inputs on small screens */
.checkout-form fieldset { display: grid; grid-template-columns: 1fr; gap: 6px 16px; align-content: start; }
.field { display: contents; }               /* label / input / message become grid items of the fieldset */
.field label { grid-column: 1; margin-top: 10px; font-weight: 500; }
.field input, .field select { grid-column: -2 / -1; width: 100%; padding: 9px 10px; border: 1px solid #bbb;
border-radius: 4px; font: inherit; background: #fff; }
.field .msg { grid-column: -2 / -1; display: none; color: #c62828; font-size: 0.85rem; }

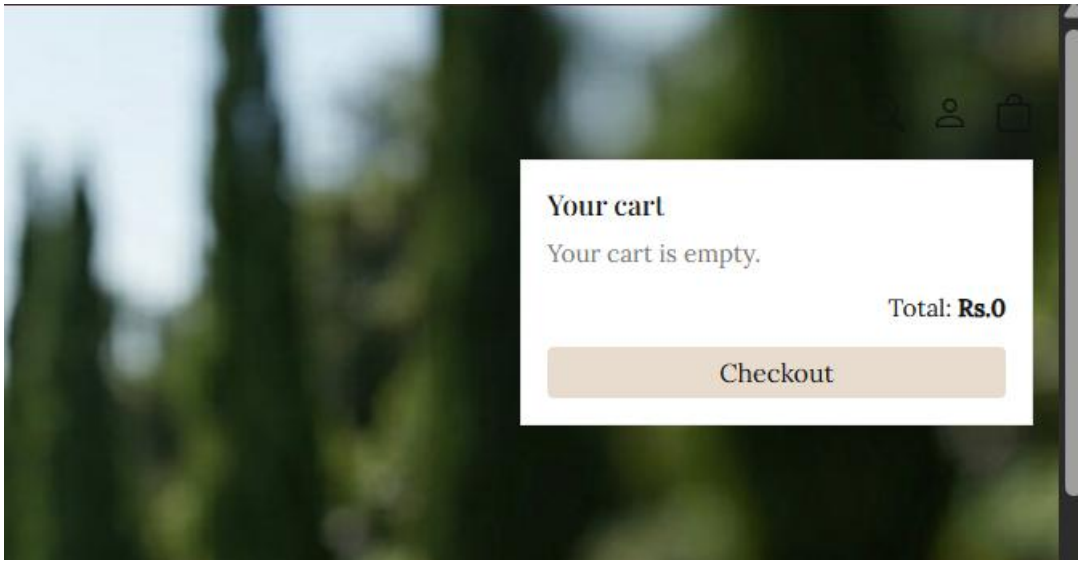
/* validation feedback only AFTER the user has interacted with the field */
.field input:user-invalid, .field select:user-invalid { border-color: #c62828; outline: 0; box-shadow: 0 0
0 2px rgba(198, 40, 40, .15); }
.field input:user-invalid ~ .msg, .field select:user-invalid ~ .msg { display: block; }
.field input:user-valid, .field select:user-valid { border-color: #2e7d32; }

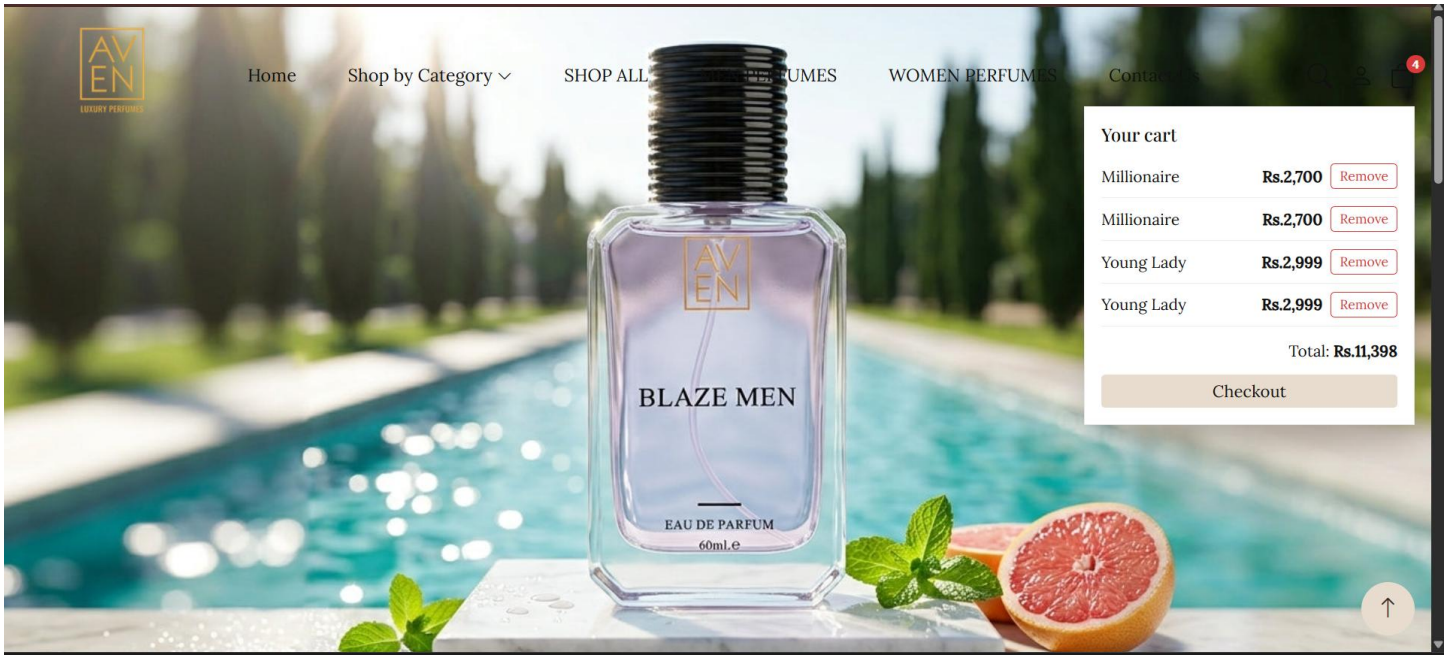
.checkout-submit { grid-column: 1 / -1; width: 100%; background: #4a2c2a; color: #fff; border: 0; border-
radius: 4px; padding: 14px; font-size: 1.05rem; }
.checkout-submit:hover { background: #351f1d; }

@media (min-width: 992px) {                  /* BP and wider */
  .checkout-form { grid-template-columns: 1fr 1fr; }                          /* fieldsets side by side */
  .checkout-form fieldset { grid-template-columns: 110px 1fr; }               /* labels in a left column */
  .field label { margin-top: 0; align-self: center; }
}
```

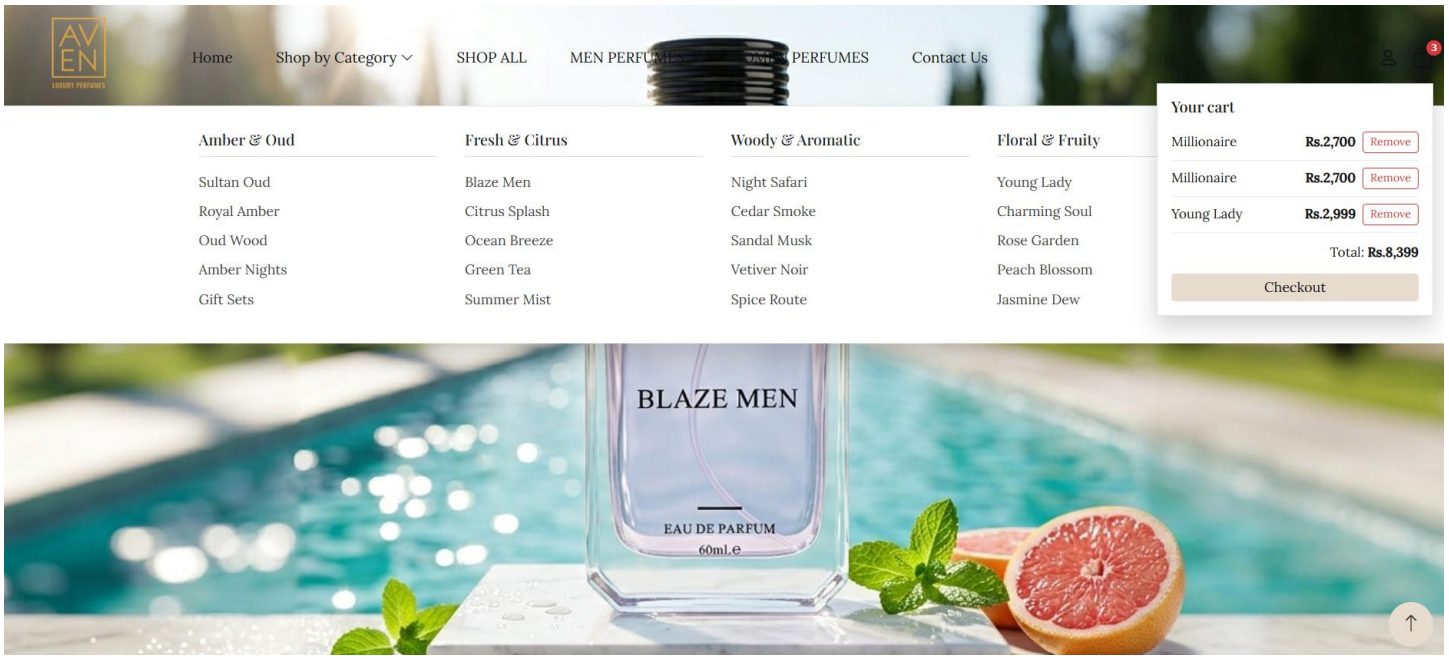
Task 3 — Cart badge and mini cart

Screenshots





Cart with 4 items — 992px



Cart with 4 items — 375px After removing one item — 375px



Cart with 4 items — 1280px

```
<div class="cart-wrap">
            <a href="checkout.html" id="cartBtn" aria-label="Cart" aria-expanded="false" aria-
controls="miniCart">
              <i class="bi bi-bag"></i><span class="cart-badge" id="cartBadge" hidden>0</span>
            </a>
            <div class="mini-cart" id="miniCart" hidden>
              <h3>Your cart</h3>
              <ul id="miniCartList"></ul>
              <p class="mini-empty" id="miniEmpty">Your cart is empty.</p>
              <p class="mini-total">Total: <strong id="miniTotal">Rs.0</strong></p>
              <a class="mini-checkout" href="checkout.html">Checkout</a>
            </div>
          </div>

<article class="product-card">
            <span class="badge-sale">ON SALE</span>
            <img src="images/Blaze_Men.png" alt="Blaze Men perfume">
            <p class="product-name">Blaze Men</p>
            <span class="price">Rs.2,400.00 PKR</span><span class="price price-old">Rs.2,900.00 PKR</span>
            <button type="button" class="btn-add">Add to Cart</button>
          </article>
```

script.js

```
/* MIDTERM TASK 3 — cart badge + mini cart (plain JavaScript) */

const N = 44;                                  // my number from the exam sheet
const MAX PER PRODUCT = (N % 4) + 2;           // a product can be added at most this many times

let cart = [];                                 // array of { name, price }

const badge     = document.getElementById('cartBadge');
const cartBtn   = document.getElementById('cartBtn');
const miniCart  = document.getElementById('miniCart');
const listEl    = document.getElementById('miniCartList');
const emptyEl   = document.getElementById('miniEmpty');
const totalEl   = document.getElementById('miniTotal');
const cards     =  document.querySelectorAll('.product-card');

// "Rs.2,400.00 PKR" -> 2400
function parsePrice(text) {
  const match = text.match(/\d[\d,]*\.?\d*/);
  return match ? parseFloat(match[0].replace(/,/g, '')) : 0;
}
function formatRs(n) {
  return 'Rs.' + n.toLocaleString('en-PK', { minimumFractionDigits: 0, maximumFractionDigits: 2 });
}
function countOf(name) {
  return cart.filter(item => item.name === name).length;
}

// refresh every part of the UI from the cart array
function render() {
  // badge
  badge.textContent = cart.length;
  badge.hidden = cart.length === 0;

  // mini cart list
  listEl.textContent = '';
  cart.forEach((item, index) => {
    const li = document.createElement('li');

    const name = document.createElement('span');
    name.className  =  'mi-name';
    name.textContent = item.name;

    const price = document.createElement('span');
    price.className = 'mi-price';
    price.textContent = formatRs(item.price);

    const remove = document.createElement('button');
```

```
    remove.type = 'button';
    remove.className = 'mi-remove';
    remove.textContent = 'Remove';
    remove.dataset.index = index;

    li.append(name, price, remove);
    listEl.appendChild(li);
  });
  emptyEl.hidden = cart.length > 0;

  // total with reduce()
  const total = cart.reduce((sum, item) => sum + item.price, 0);
  totalEl.textContent = formatRs(total);

  // Add to Cart buttons: limit reached?
  cards.forEach(card => {
    const btn = card.querySelector('.btn-add');
    const name = card.querySelector('.product-name').textContent.trim();
    const reached = countOf(name) >= MAX_PER_PRODUCT;
    btn.disabled = reached;
    btn.textContent = reached ? 'Limit reached' : 'Add to Cart';
  });
}

// Add to Cart (name and price are read from the card itself)
cards.forEach(card => {
  card.querySelector('.btn-add').addEventListener('click', () => {
    const name  = card.querySelector('.product-name').textContent.trim();
    const price = parsePrice(card.querySelector('.price:not(.price-old)').textContent);
    if (countOf(name) >= MAX_PER_PRODUCT) return;
    cart.push({ name, price });
    render();
  });
});

// Remove button (event delegation)
listEl.addEventListener('click', e => {
  const btn = e.target.closest('.mi-remove');
  if (!btn) return;
  cart.splice(Number(btn.dataset.index), 1);
  render();
});

// cart icon toggles the mini cart (without JS it simply links to checkout.html)
cartBtn.addEventListener('click', e => {
  e.preventDefault();
  const open = miniCart.hidden;             // currently closed -> open it
  miniCart.hidden = !open;
  cartBtn.setAttribute('aria-expanded', String(open));
});

render();
```

CSS

```
/* ========================================================================
   MIDTERM TASK 3 — cart badge, mini cart, Add to Cart buttons
   ======================================================================== */
.cart-wrap { position: relative; }
.icons a#cartBtn { position: relative; display: inline-block; }
.cart-badge {
  position: absolute; top: -8px; right: -10px;
  min-width: 18px; height: 18px; padding: 0 5px; border-radius: 9px;
  background: #d93a3a; color: #fff; font: 700 11px/18px sans-serif; text-align: center;
}
.cart-badge[hidden], .mini-cart[hidden] { display: none; }
.mini-cart {
  position: absolute; top: calc(100% + 14px); right: 0; z-index: 2100;
  width: min(320px, calc(100vw - 32px)); background: #fff; color: #111;
  border: 1px solid #e6e6e6; box-shadow: 0 12px 28px rgba(0, 0, 0, .18); padding: 16px;
  font-size: 0.95rem;
```

```
}
.mini-cart h3 { font-family: 'Playfair Display', Georgia, serif; font-size: 1.1rem; margin: 0 0 10px; }
.mini-cart ul { list-style: none; margin: 0; padding: 0; max-height: 260px; overflow-y: auto; }
.mini-cart li { display: flex; justify-content: space-between; align-items: center; gap: 8px; padding: 8px
0; border-bottom: 1px solid #eee; }
.mini-cart .mi-name { flex: 1; }
.mini-cart .mi-price { font-weight: 700; white-space: nowrap; }
.mini-cart .mi-remove { border: 1px solid #d93a3a; background: #fff; color: #d93a3a; border-radius: 4px;
font-size: 0.8rem; padding: 2px 8px; }
.mini-cart .mi-remove:hover { background: #d93a3a; color: #fff; }
.mini-total { margin: 12px 0; text-align: right; }
.mini-empty { color: #777; margin: 0; }
.mini-checkout { display: block; text-align: center; background: #e8dccf; color: #111 !important; padding:
8px; border-radius: 4px; text-decoration: none; font-size: 1rem !important; }
.btn-add { margin-top: 10px; width: 100%; border: 1px solid #111; background: #fff; color: #111; border-
radius: 4px; padding: 7px 0; font-size: 0.85rem; }
.btn-add:hover:not(:disabled) { background: #111; color: #fff; }
.btn-add:disabled { background: #eee; border-color: #ccc; color: #888; cursor: not-allowed; }/*
========================================================================
   MIDTERM TASK 3 — cart badge, mini cart, Add to Cart buttons
   ======================================================================== */
.cart-wrap { position: relative; }
.icons a#cartBtn { position: relative; display: inline-block; }
.cart-badge {
  position: absolute; top: -8px; right: -10px;
  min-width: 18px; height: 18px; padding: 0 5px; border-radius: 9px;
  background: #d93a3a; color: #fff; font: 700 11px/18px sans-serif; text-align: center;
}
.cart-badge[hidden], .mini-cart[hidden] { display: none; }
.mini-cart {
  position: absolute; top: calc(100% + 14px); right: 0; z-index: 2100;
  width: min(320px, calc(100vw - 32px)); background: #fff; color: #111;
  border: 1px solid #e6e6e6; box-shadow: 0 12px 28px rgba(0, 0, 0, .18); padding: 16px;
  font-size: 0.95rem;
}
.mini-cart h3 { font-family: 'Playfair Display', Georgia, serif; font-size: 1.1rem; margin: 0 0 10px; }
.mini-cart ul { list-style: none; margin: 0; padding: 0; max-height: 260px; overflow-y: auto; }
.mini-cart li { display: flex; justify-content: space-between; align-items: center; gap: 8px; padding: 8px
0; border-bottom: 1px solid #eee; }
.mini-cart .mi-name { flex: 1; }
.mini-cart .mi-price { font-weight: 700; white-space: nowrap; }
.mini-cart .mi-remove { border: 1px solid #d93a3a; background: #fff; color: #d93a3a; border-radius: 4px;
font-size: 0.8rem; padding: 2px 8px; }
.mini-cart .mi-remove:hover { background: #d93a3a; color: #fff; }
.mini-total { margin: 12px 0; text-align: right; }
.mini-empty { color: #777; margin: 0; }
.mini-checkout { display: block; text-align: center; background: #e8dccf; color: #111 !important; padding:
8px; border-radius: 4px; text-decoration: none; font-size: 1rem !important; }
.btn-add { margin-top: 10px; width: 100%; border: 1px solid #111; background: #fff; color: #111; border-
radius: 4px; padding: 7px 0; font-size: 0.85rem; }
.btn-add:hover:not(:disabled) { background: #111; color: #fff; }
.btn-add:disabled { background: #eee; border-color: #ccc; color: #888; cursor: not-allowed; }

@media (max-width: 991.98px) {               /* below BP: mini cart spans the screen width, just under the
navbar icons */
  .mini-cart { position: fixed; top: 78px; left: 16px; right: 16px; width: auto; }
}
```